

DEPARTMENT OF MATHEMATICS & STATISTICS
MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: (II)

Problem Set 0

(Review Exercises from MSO 205)

[1] Let $\Omega = \{1, 2, 3, 4\}$. Check whether the following classes of sets are σ -fields or not:

$$F_1 = \{\emptyset, \{1, 2\}, \{3, 4\}\}$$

$$F_2 = \{\emptyset, \Omega, \{1\}, \{2, 3, 4\}, \{1, 2\}, \{3, 4\}\}$$

$$F_3 = \{\emptyset, \Omega, \{1\}, \{2\}, \{1, 2\}, \{3, 4\}, \{2, 3, 4\}, \{1, 3, 4\}\}$$

[2] Let F be a σ -field of subsets of the sample space Ω and let $A \in F$ be fixed. Verify whether $F_A = \{C : C = A \cap B, B \in F\}$ is a σ -field of subsets of A or not.

[3] Let $\Omega = \{0, 1, 2, \dots\}$. In each of the following cases check if P is a probability measure, where, for an event A ,

$$(a) P(A) = \sum_{x \in A} \frac{e^{-\lambda} \lambda^x}{x!}, \lambda > 0.$$

$$(b) P(A) = \sum_{x \in A} p(1-p)^x, 0 < p < 1.$$

$$(c) P(A) = \begin{cases} 1 & \text{if the number of elements in } A \text{ is finite} \\ 0 & \text{otherwise} \end{cases}$$

In cases where P is a probability measure, find $P(E), P(F), P(G), P(E \cap F), P(E \cup F), P(F \cup G), P(E \cap G)$, and $P(F \cap G)$, if $E = \{x \in \Omega : x > 2\}$, $F = \{x \in \Omega : 0 < x < 3\}$ and $G = \{x \in \Omega : 3 < x < 6\}$.

[4] Let $\Omega = \mathbb{R}$. In each of the following cases check if P is a probability measure, if for an interval I ,

$$(a) P(I) = \int_I \frac{1}{2} e^{-|x|} dx$$

$$(b) P(I) = \int_I \frac{1}{\pi} \frac{1}{1+x^2} dx$$

$$(c) P(I) = \begin{cases} 0 & \text{if } I \subset (-\infty, 1) \\ \int_I \frac{dx}{2} & \text{if } I \subset [1, \infty) \end{cases}$$

$$(d) P(I) = \begin{cases} 1 & \text{if length of } I \text{ is finite} \\ 0 & \text{otherwise} \end{cases}$$

In cases where P is a probability measure, find $P(x : x \geq 0)$.

[5] A coin is tossed until for the first time the same result appear twice in succession. To an outcome requiring n tosses assign a probability 2^{-n} . Describe the sample space. Evaluate the probability of the following events:

- (a) A = the experiment ends before the 6th toss;
- (b) B = an even number of tosses are required;
- (c) $A \cup B, A \cap B, A \cap B^c, A^c \cap B^c, A^c \cap B$

[6] A system composed of n separate components is said to be a parallel system if it functions when at least one of the components functions. For such a system, if component i , independent of other components, functions with probability $p_i, i=1(1)n$, what is the probability that the system functions?

[7] Let $A_1, A_2, \dots, A_n$ be n independent events. Show that

$$P\left(\bigcap_{i=1}^n A_i^c\right) \leq \exp\left(-\sum_{i=1}^n P(A_i)\right)$$

[8] We say that B carries negative information about event A if $P(A|B) < P(A)$. Let A, B, C be three events such that B carries negative information about A and C carries negative information about B . Prove or disprove that C carries negative information about A .

[9] Let $\Omega = \{1, 2, 3, 4\}$ and $\mathfrak{F} = \{\phi, \Omega, \{1\}, \{2\}, \{1, 2\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}$. Verify if $\mathfrak{F}$ is a σ -field defined on the subsets of Ω . Prove or disprove $X(\omega) = \omega + 2, \forall \omega \in \Omega$ is a random variable with respect to the given $(\Omega, \mathfrak{F})$.

[10] Let X be a random variable defined on $(\Omega, \mathfrak{F}, P)$. Show that the following are also random variables; (a) $|X|$, (b) X^2 and (c) $\sqrt{X}$, given that $\{X < 0\} = \phi$.

[11] Let $\Omega = [0, 1]$ and $\mathfrak{F}$ be the σ -field of subsets of Ω . Define X on Ω as follows:

$$X(\omega) = \begin{cases} \omega & \text{if } 0 \leq \omega \leq 1/2 \\ \omega - 1/2 & \text{if } 1/2 < \omega \leq 1 \end{cases}$$

Show that X defined above is a random variable.

[12] Let

$$F(x) = \begin{cases} 0 & \text{if } x < -1 \\ (x+2)/4 & \text{if } -1 \leq x < 1 \\ 1 & \text{if } x \geq 1. \end{cases}$$

Show that $F(\cdot)$ is a distribution function. Sketch the graph of $F(x)$ and compute (a) $P(-1/2 < X \leq 1/2)$, (b) $P(X = 0)$, (c) $P(X = 1)$ and (d) $P(-1 \leq X < 1)$. Further, obtain the decomposition $F(x) = \alpha F_d(x) + (1 - \alpha)F_c(x)$, where $F_d(x)$ and $F_c(x)$ are purely discrete and purely continuous distribution functions, respectively.

- [13] Suppose F_X is the distribution function of a random variable X . Determine the distribution function of (a) X^+ , (b) X^- , (c) $|X|$ and (d) X^2 , where

$$X^+ = \begin{cases} X & \text{if } X \geq 0 \\ 0 & \text{if } X < 0 \end{cases} \quad \text{and} \quad X^- = \begin{cases} -X & \text{if } X < 0 \\ 0 & \text{if } X \geq 0 \end{cases}$$

- [14] Check whether the function given by

$$f(x) = \frac{x+2}{25} \text{ for } x = 1, 2, 3, 4, 5, \text{ can be a probability mass function.}$$

- [15] Show that the function $f(x) = \frac{1}{2}e^{-|x|}$, $-\infty < x < \infty$, is a probability density function.

- [16] What is the smallest value of k in Chebyshev's inequality for which the probability that an r.v. X with mean μ and variance σ^2 will take on a value between $\mu - k\sigma$ and $\mu + k\sigma$ is at least (a) 0.95 and (b) 0.99?

- [17] Let X be an r.v. with mean μ . If it is known $P(X \leq 0) = 0$, prove or disprove that $P(X \geq 2\mu) \leq 0.5$.

- [18] Obtain the moment generating function (m.g.f.) of an r.v. with p.d.f.

$$f(x) = \frac{1}{2}e^{-|x|} \text{ for } -\infty < x < \infty$$

- [19] If the probability that an applicant for a driving license will pass the road test on any given try is 0.75, what is the probability that an applicant will pass the test in his fourth attempt?

- [20] In a certain city, the daily consumption of electric power in millions of kilo-watt-hours can be treated as a random variable having a gamma distribution with shape 3 and scale 2. If the power plant of this city has a daily capacity of 12 million kilowatt-hours, what is the probability that this power supply will be inadequate on any given day?

- [21] The mileage (in thousands of kms) that car owners get with a certain kind of radial tyre is a random variable having an exponential distribution with $q = 40$. Find the probabilities that one of these tyres will last (a) at least 20,000 kms; and (b) at most 30,000 kms.

[22] Let $X \sim N(\mu, \sigma^2)$. If $[X \leq 0] = 0.5$, $P[-1.96 \leq X \leq 1.96] = 0.95$, determine μ, σ^2 .

[23] To pass a paper of 100 marks, an examinee requires to score at least 33. To clear the paper in second division, he requires at least 51 but less than 60; and for first division he needs at least 60. If the distribution of marks received in the examination is normal and it is known that out of 10000 candidates, 22 per cent failed and 1515 passed in the first division, find the number of candidates passing in second division.
(Use $\Phi(0.43) = 0.673$, $\Phi(0.77) = 0.779$, $\Phi(1.03) = 0.853$)

[24] Let $X_1, \dots, X_n$ be a random sample from $U(0, 1)$. Let $Y_n = \min(X_1, \dots, X_n)$ and $Z_n = \max(X_1, \dots, X_n)$. Show that (a) $\sqrt{Y_n} \xrightarrow{p} 0$, (b) $Z_n^2 \xrightarrow{p} 1$ and (c) $Y_n^2 Z_n^2 \xrightarrow{p} 0$.

[25] Let $X_1, \dots, X_n$ be i.i.d. $N(0, 1)$. Show that $\bar{X}_n / S_n \xrightarrow{p} 0$, where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ and $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.

[26] Let $\{X_n\}$ be a sequence of i.i.d. random variables with $U(0, 1)$ distribution and $Z_n = \left\{ \prod_{i=1}^n X_i \right\}^{1/n}$. Show that $Z_n \xrightarrow{p} 1/e$.

[27] Let $\{X_n\}$ be a sequence of $N\left(\frac{1}{n}, 1 - \frac{1}{n}\right)$ variables. Show that $X_n \xrightarrow{L} Z \sim N(0, 1)$.

[28] Let $X_1, X_2, \dots$ be i.i.d. r.v.'s with $E[X_i] = \mu$, $\text{Var}[X_i] = \sigma^2$ and $E[(X_i - \mu)^4] = \sigma^4 + 1$.
(a) Does the WLLN's hold for the r.v.'s $X_1^2, X_2^2, \dots$?

(b) Find $\lim_{n \rightarrow \infty} \left[\sigma^2 - \frac{1}{\sqrt{n}} \leq \frac{(X_1 - \mu)^2 + \dots + (X_n - \mu)^2}{n} \leq \sigma^2 + \frac{1}{\sqrt{n}} \right]$.

[29] Let $X_1, \dots, X_{100}$ be i.i.d. from Poisson distribution $P(3)$ distribution and let $Y = \sum_{i=1}^{100} X_i$. Find an approximate value of $P(100 \leq Y \leq 200)$.

[30] The p.d.f. of X_n is given by

$$f_n = \begin{cases} \frac{1}{\sqrt{n}} e^{-x} x^{n-1} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the limiting distribution of $Y_n = X_n / n$.